ONE-PAGER · BRIEF 12 · ACCESS · AFFORDABILITY · AVAILABILITY

Relocatable clean power for a hospital on the move

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Focus region Enugu, Nigeria | **Worked example** National Orthopaedic Hospital Enugu | **Version** POC summary · Brief 12

The National Orthopaedic Hospital Enugu gets about 6 h of grid a day against about 1,550 kW of need, diesel costs ₦3,277/L (up 86 % in a year), and the hospital moves building in about five years, so nothing can be fixed to today's walls [1][4][11].

THE THREE A'S IN ENUGU

	TODAY IN ENUGU (SOURCED)	WHAT GRIDRESCUE DOES [13]
ACCESS	About 6 h of grid a day against 1,550 kW [1]; the state publishes feeder supply hours and class [5].	Reads any site from open data: 9 substations within 50 km, 5.48 peak-sun hours confirmed by four solar sources, Band A ₦160/kWh.
AFFORDABILITY	Diesel ₦3,277/L, up 86 % in a year [4]; teaching-hospital bills three to four times higher since 2024 [3]; little upfront capital [11].	1,670 kWp ground-mounted solar, 5,020 kWh LFP in 3 battery containers (5 in all), 500 kVA UPS, generator last; capex ₦3.85 bn. ₦0 upfront: ₦299.2 M/month of modelled diesel becomes a ₦69.1 M/month fee, saving from month one if diesel is above ₦117.9 M/month; payback 1.6 years; 10 years ₦15.9 bn vs ₦37.6 bn on diesel.
AVAILABILITY	Grid drops without warning; switchovers damage equipment [1]; theatres must never lose power [11]; the hospital moves in about five years [1].	4 of 4 stress tests pass; 100 % survival of any outage up to 24 h, whatever hour it starts; 866,678 L diesel (83 %) and 2,246 t CO_2 (net) avoided a year; move 4 km north in about 3 weeks for ₦678 M, 88 % of the investment reused.

WHAT WAS BUILT · FOUR STEPS OF THE RUNNING DEMO

1 MAP AND READ	2 MOVE AND LOCK	3 SIMULATE	4 IMPACT AND 3D
Pin the hospital; a sun lens reads NASA POWER, PVGIS, PVWatts and the Global Solar Atlas, plus wind and water. Substations within 50/100 km, supply hours and tariff.	Areas within 12 km scored on grid access (OpenStreetMap) and distance; user locks the new site. Relocation plan answers Brief 12's five starter questions.	The engine simulates every storage family hour by hour. Without a generator: LFP ₦4.65 bn vs hydrogen ₦8.18 bn (almost twice), both 100 % theatres; grid-first UPS fallback.	Diesel and CO_2 avoided, Scope 1/2 ESG lines; carbon credits: potential, valued only after verification and issuance. 3D: hospital through a 24 h collapse; rail crew 45 dB vs 68 dB.

OUTAGES AND EXTENSIONS

Built: 12 h evening outage, 24 h national grid collapse, worst solar month, 18 h outage, all at site level.

Extension: feeder or substation trips, load shedding and band cuts, frequency collapse (grid); one feeder, district or state market, storms, floods, heat waves (location); cyber-security outages that cut power or controls.

Same engine, more use cases: night rail crew (built), clinic cold storage (Brief 01), surplus solar to anchor customers [10], EV charging, telecom towers, feeder planning. Vendor-neutral; we sell no hardware.

ESG before purchase: Nigeria's Financial Reporting Council is rolling out the ISSB standards (IFRS S1/S2), which ask for Scope 1 and 2 emissions [20]; GridRescue produces the site's Scope 1/2 and E · S · G report before anything is bought.

JUDGING FIT [11] Impact 25 %: 4,992 at-risk theatre-hours a year → 0, 2,246 t CO_2 (net) avoided a year · Innovation 25 %: relocatable design chosen by simulation · Sustainability 25 %: 88 % reused, ₦21.8 bn saved over 10 years · Feasibility 15 %: 4/4 stress tests on published data · Scalability 10 %: global data, only a tariff table per country.

OUR TEAM SUBMITS Demo video (under 5 min), pitch video (under 5 min), 5-slide pitch deck, POC summary, disclosure statement. Figures marked [13] are engine results for the worked example; to be confirmed with the hospital's load data.

KEY REFERENCES

[1] National Orthopaedic Hospital Enugu, head engineer, "Problem statement and background shared with the Global Energy Hackathon organizers," 2026 (reproduced in Problem Brief 12).

[3] AllAfrica, "Nigeria: Electricity Tariff... Life-Saving Services Threatened in Teaching Hospitals," 2024. [Online]. Available: https://allafrica.com/stories/202406100036.html

[4] Legit.ng, "NBS: Diesel Price Rises 86.40% To N3,277/Litre In May 2026," 2026. [Online]. Available: https://www.legit.ng/business-economy/energy/1716140-nbs-diesel-price-rises-8640-n3277litre-2026/

[5] Nairametrics, "Enugu slashes electricity tariff for Band A customers to N160/kWh effective August 1," 2025. [Online]. Available: https://nairametrics.com/2025/07/21/enugu-slashes-electricity-tariff-for-band-a-customers-to-n160-kwh-effective-august-1/

[10] ThisDay, "EU, Enugu Govt to Spend N4.3bn on Solar Energy in Healthcare Centres," 2024. [Online]. Available: https://www.thisdaylive.com/2024/11/20/eu-enugu-govt-to-spend-n4-3bn-on-solar-energy-in-healthcare-centres/

[11] Global Energy Hackathon, "Problem Brief 12: Relocatable solar for a hospital on the move," Clean Energy Solutions track, hackathon.anneru.com (success metrics, constraints and starter questions).

[13] GridRescue AI, "GridRescue engine, Enugu worked example," output of the running proof-of-concept application for the National Orthopaedic Hospital Enugu site (engine results; to be confirmed with the hospital's load data).

[20] Financial Reporting Council of Nigeria, "Roadmap Report for the Adoption of IFRS Sustainability Disclosure Standards in Nigeria (Amended 2026)," 2026. [Online]. Available: https://frcnigeria.gov.ng/wp-content/uploads/2026/03/Roadmap-Report-for-the-Adoption-of-IFRS-Sustainability-Disclosure-Standards-in-Nigeria-Amended-2026-.pdf